



ASIR

HARDWARE FUNDAMENTALS

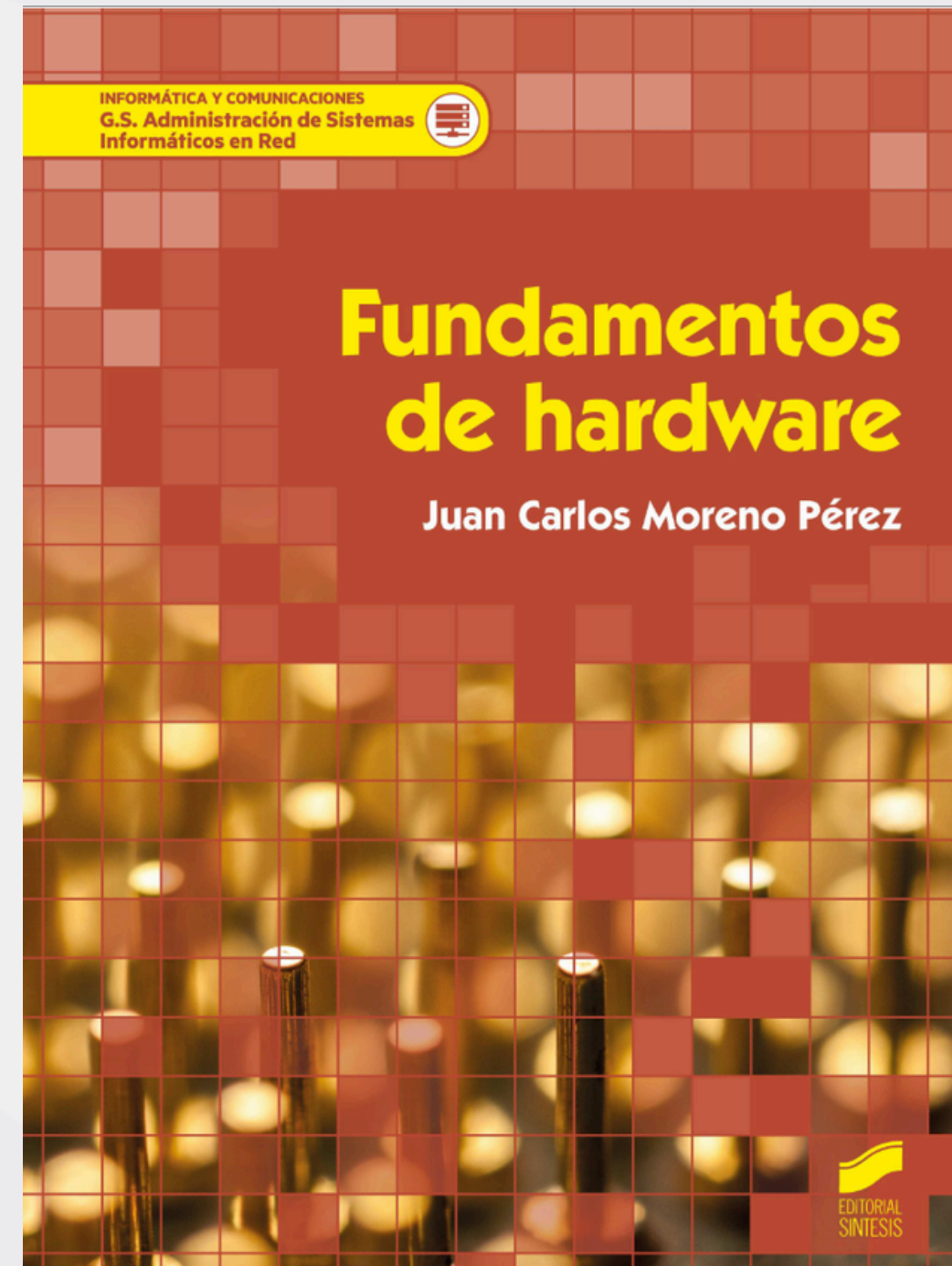
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BOOK



MATERIALS

GitHub: <https://github.com/BegoRodriguez/DAW-Web-Interfaces-Design>

Dropbox:

https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0

- Link to contents of DAW - Web Interfaces Design:
https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0

Pinned

Customize your pins

 **DAM-Sistemas-informaticos** Public

Materials para el módulo Sistemas Informáticos del IES Ribera de Castilla de Primero de DAM

 **SMR-Innovacion** Public

Materials para el módulo Innovación aplicada al sector productivo del IES Ribera de Castilla de Primero de SMR

 **DAW-Web-Interfaces-Design** Public

Materials for Web Interfaces Design in IES Ribera de Castilla

 **Recursos** Public

Compilación de recursos gratuitos de programación, bases de datos y tecnología en general.

 8  3

TEAMS

I have created a classroom team for communication and project and activities delivery on Teams.

This is the code:

```
icp0asz
```

COMPUTER ARCHITECTURES: FUNCTIONAL ELEMENTS OF A COMPUTER

Objectives:

- Explore computing from the perspective of computer hardware and systems.
- Understand the classical Von Neumann architecture, including its functional units and subsystems.
- Learn about the architectures currently used in mobile devices.
- Understand the concept of System-on-Chip (SoC).
- Identify emerging processor architectures, including neuromorphic chips and AI-oriented architectures.

COMPUTER ARCHITECTURES: FUNCTIONAL ELEMENTS OF A COMPUTER (II)

- Introduction
- Computer Architecture. Functional Units and Subsystems
 - Memory
 - Input/Output Subsystem
 - The ALU (Arithmetic Logic Unit)
 - The Control Unit
- SoCs. Mobile Processors
- New Microprocessor Architectures: Neuromorphic Chips
- Artificial Intelligence in Processors

THINK ABOUT IT FOR A MOMENT...

Computers are part of everything we do. We use them without even noticing: surfing the web, writing, emailing, gaming, watching content. So let's start with a simple but powerful question: what is a computer, really? An appliance? A toy? A typewriter? Something more?



AUTOEVALUATION

A computer is...

... a machine for writing documents

... a calculator

... a general purpose machine that works with given data and provides results

... a machine to see TV

AUTOEVALUATION (II)

A computer is...

... a machine for writing documents

... a calculator

... **a general purpose machine that works with given data and provides results**

... a machine to see TV

COMPUTER DEFINITION

A computer is an electronic, programmable device capable of receiving data as input, processing and storing it according to a set of instructions, and producing information as output.

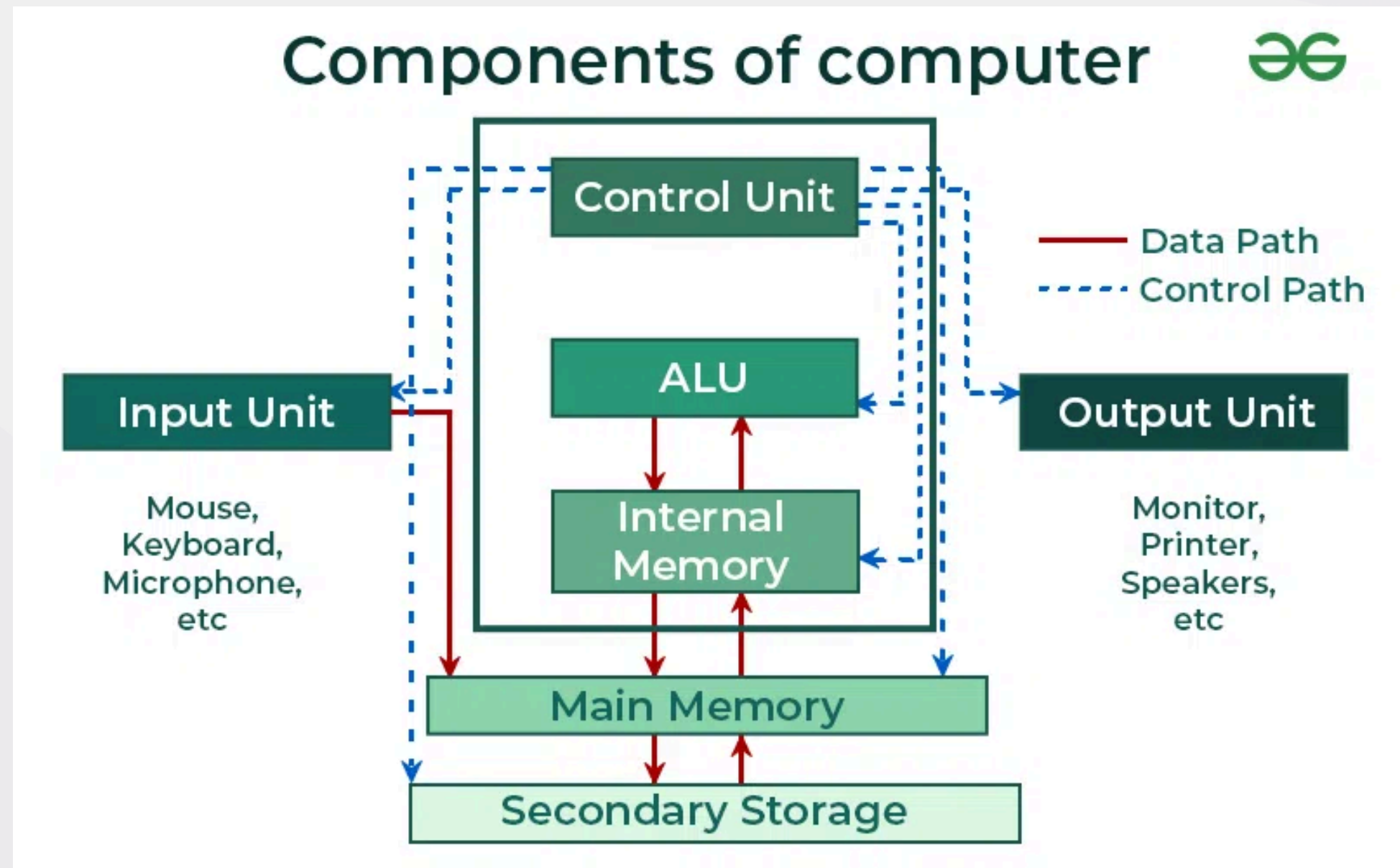
A computer system and is composed of two parts: hardware (the physical elements) and software (the instructions that tell the hardware what it has to do, how to solve the problem). Sometimes, some authors include two parts more: the user and the data.

ACTIVITY: COMPUTER HISTORY

Write a short article about the origins of computer science and the development of the first computers. Include references to early machines such as the Pascaline, Colossus, ENIAC, and the Intel 8086, among others.

COMPUTER ARCHITECTURE

Computer architecture is a theoretical model that describes the structure, components, and organization of a computer system



VON NEUMANN ARCHITECTURE

Von Neumann architecture is an abstract model created in 1946 that describes a computer architecture.

In Von Neumann architecture, the main elements of a computer are:

- CPU (Central Process Unit).
- Memory unit.
- Buses.
- I/O (Input/Output to external devices).

VON NEUMANN ARCHITECTURE (II)

Von Neumann architecture works as described:

- Components are connected by buses.
- It can execute elemental instructions (machine code).
- The instructions are stored in the memory unit. Also, the data has to be stored in memory unit.
- Both instructions and data could be provided by input devices or generated by other programs (compiler, linker, etc.).
- The CPU reads an instruction and the control unit generates each signal to perform the instruction.
 - If it is required, CPU can perform arithmetic-logic operations.
- The CPU results can be stored on the memory unit, or it can be sent to output devices.

VON NEUMANN ARCHITECTURE (III)

Over the years, processors and memory have advanced dramatically to meet growing computational needs → Moore's law

Modern CPUs achieve remarkable performance, and memory systems provide massive storage capacity.

This evolution has led to highly specialized components, enabling microprocessors to integrate billions of transistors on a single chip.

Nowadays, there are several improvements of this architecture, including features like mapped I/O (memory unit and I/O devices are treated like the same memory).

If you want to try how Von Neumann architecture works, you can use this website:

<https://lab.xitrus.es/VonNeumann/>

FUNCTIONAL ELEMENTS OF A COMPUTER

In this point, we are going to describe functional elements of a computer that implements Von Neumann architecture.

CPU(CENTRAL PROCESS UNIT)

The CPU is the core of the computer. CPU read instructions from memory and execute them, doing the required operations. It is composed of:

- **Registers:** they are little and fast memories. They are used as temporal storage when an instruction is being executed or to store temporally results of operations. The most important registers are:
 - **Program counter (PC):** an incremental counter that stores the memory address of the next instruction that is going to be executed.
 - **Memory address register (MAR):** it stores the address of a block of memory for reading from or writing to.

CPU (CENTRAL PROCESS UNIT) (II)

- **Registers (II):**

- **Memory data register (MDR):** a two-way register that stores data fetched from memory (and ready for the CPU to process) or data waiting to be stored in memory.
- **Instruction register (IR):** it is a temporal register to store the instruction that has just been fetched from memory.
- **General purpose registers:** those registers are used to store information temporally, for example to store operands of an addition or to store the result of an arithmetic-logic operation.
- **Flags:** These are very simple records. They can only be in two states: "On" or "Off". They are used to indicate the status of an operation. Example: flag Z is set to 1 if the operation has resulted in 0, N is set to 1 if the result is negative.

CPU(CENTRAL PROCESS UNIT) (III)

- **Internal bus:** is an internal bus that connect registers, control unit and arithmetic-logic unit.
- **Control unit (UC):** it decides what to do in each moment. Its components are:
 - **Decoder:** its function is to decode instructions and prepare all the signals to execute the program properly.
 - For instance, an instruction can be “ADD” (which adds up two numbers that are in memory). When it comes to executing, the decoder generates a lot of small signals such as reading the first number of the memory, taking it to the ULA, reading the second number, taking it to the ULA, perform the operation (adding) and saving the result in memory. In this case, an instruction has generated six signals 1.
 - **Clock:** is the system that sets the pace at which things are done. It's like the conductor's baton. It allows all system components to operate perfectly together.

CPU(CENTRAL PROCESS UNIT) (III)

- **Arithmetic-Logic Unit (ULA):** it is a unit designed to perform arithmetic operations (addition, subtraction, multiplication, etc.) and logic operations (not, and, or, ...). It's the worker.

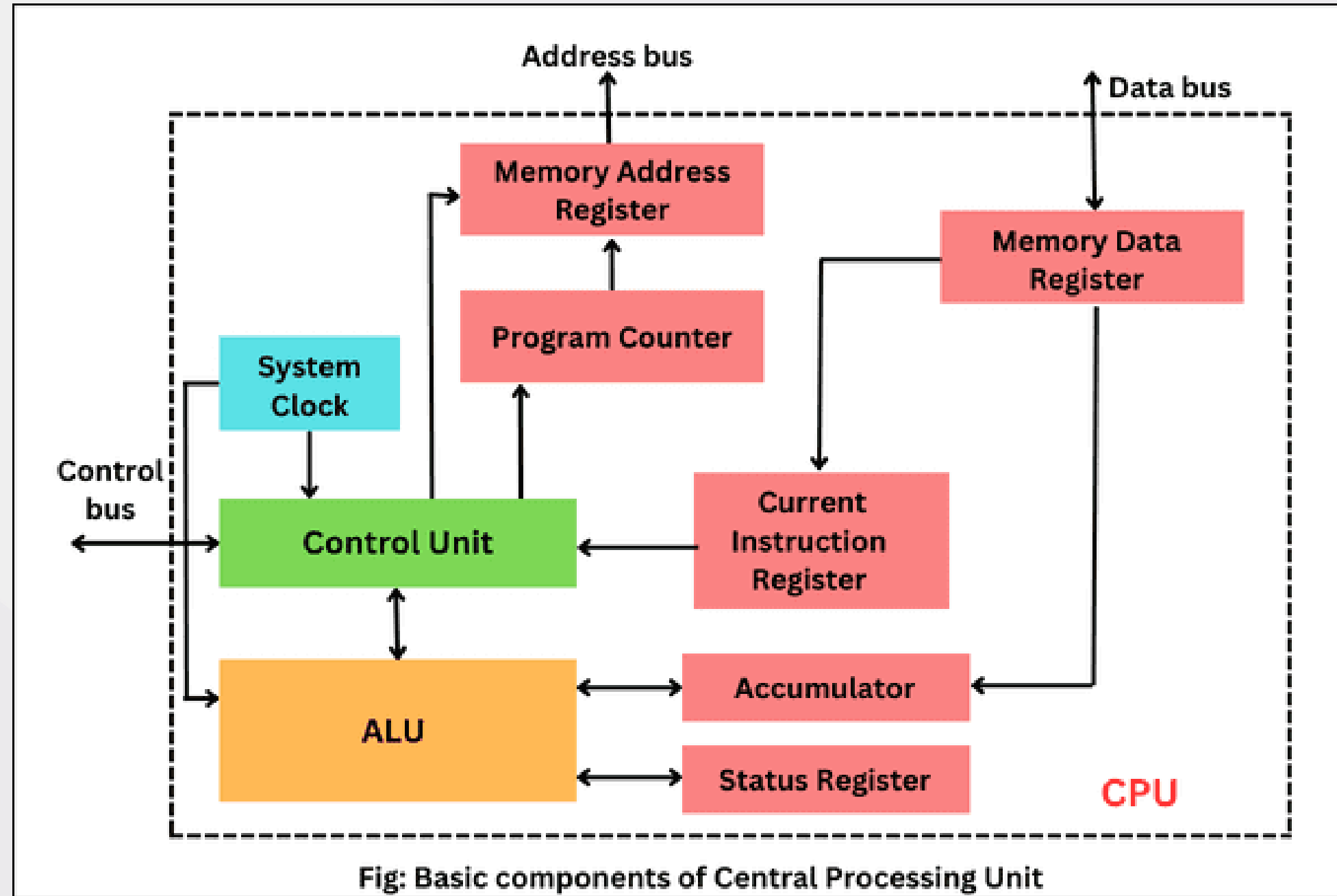
To execute an instruction, the CPU follow those steps:

- Use the Program Counter to fetch the new instruction from memory.
- Decode the instruction and fetch data from memory if it is required.
- Perform the instruction and store data in an internal register or memory.

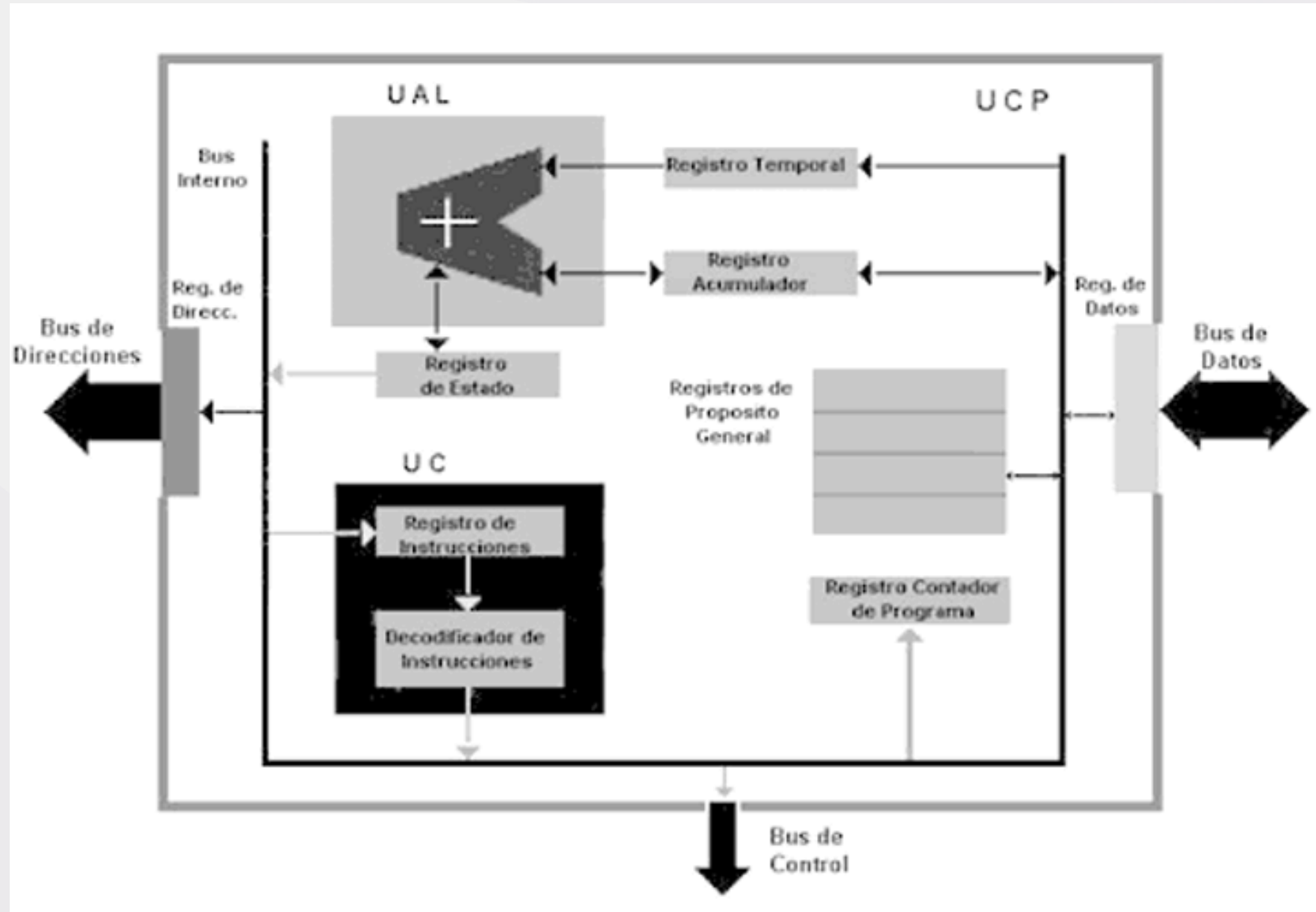
The instruction cycle will be detailed in later in the section Instruction cycle.

to measure the speed of a CPU, we use frequency units that indicate the number of times a thing repeats itself per unit of time. In general, we use the Hz (or its multiples) that indicates the number of operations that can be performed in a second.

CPU (CENTRAL PROCESS UNIT) (IV)



CPU(CENTRAL PROCESS UNIT) (V)



¿ANY QUESTION?